WANKOE Line Design Report

Limestone processing line — complete design, calculation and verification record

NOEZYS · AI Innovation Lab — SAMPLE (chapters 0 and 8 of 10), 2026-08-14 — confidential

0 Executive summary

The whole line on one page — what is sold, what is proven, what remains open.

The WANKOE line converts a single quarry feed into four sold products. At the configuration adopted on 2026-08-14 — the redesigned zone 1.3 operating in two modes behind a grits diverter, dedicated AgLime campaigns in zone 1.2, and the impactor at its low-speed setting — the annual plan serves **every market exactly**:

Product	Sold (t/y)	Status	Zone utilization
KFS 20/35 (kiln, firm)	85 000	envelope compliant	zone 1.1 — 71.2 %
FeedLime grits 2/4 (firm)	40 000	D6 compliant	zone 1.3 — 60.2 % (2 466 h mode G + 1 144 h mode F)
FeedLime fines 0/1.5 (objective)	60 000	96.6 % < 1.7 mm	
AgLime 0/1.7	135 000	98.4 % < 1.7 mm	zone 1.2 — 40.5 %

The single open cost is the disposal of surplus crushed fines: **13 816 t/y of 0/20 still go to landfill** at today's measured feed. The governing indicator is the *KFS Yield*, defined by the client and computed on every engine run:

$$Y_{KFS} = \frac{\dot{m}^{wet}_{KFS}}{\dot{m}^{wet}_{pivot}} \qquad Y^{req}_{KFS} = \frac{T_{KFS}}{T_{KFS} + D_{0/20}} \tag{0.1}$$

Today the line realizes $Y_{KFS} = 24.88\,\%$ against a zero-landfill requirement of $25.93\,\%$. Closing the 1.05-point gap is the object of the adopted quarry works specification (chapter 8): at the target inlet curve the balance closes exactly — zero landfill with a 20 % AgLime-market buffer held open as the shock absorber.

> **Confidence statement.** The zone-1.1 calculation chain passed a six-action internal verification program: blind independent recalculation (full concordance, 25 scalars + 29-point curve), literature anchors (7/7, sources cited in-test), 580-scenario invariant sweep, a Monte-Carlo uncertainty band (Y_{KFS} = 24.3 % [P10 22.4 – P90 26.0]), numerical-grid convergence (adopted ×2 grid), and a two-lens adversarial formula audit (zero sign or unit errors; three boundary-condition fixes). Meta-score **94 %** — the internal ceiling; the remaining points require the vendor gradation test, drop-weight and sieve tests, repeat belt-cuts, and the design-curve reconciliation with the engineering contractor.

The redesign in one paragraph

The as-built zone 1.3 ground 2.83 tonnes of fines for every tonne of grits and failed the adopted grits specification (15.4 % below 2 mm against a 15 % limit). The adopted C1 circuit — two-stage smooth rolls with immediate product relief on a double-deck screen pair — cuts the ratio to 0.84, passes the specification, and, operated in two modes behind the grits diverter DV.GF, decouples the grits and fines order books entirely. Purchase specifications: RC.1 32 t/h; RC.2 two units of 22 t/h each, gap range down to 1.5 mm (mode F), all [H] coefficients pending the vendor gradation test.

8 Quarry works specification

The adopted inlet-PSD target: zero landfill with a 20 % AgLime-market margin.

8.1 Definition of the target

The firm and objective sales (KFS 85 kt, grits 40 kt, fines 60 kt) fix the downstream demand for crushed 0/20; the AgLime market absorbs the remainder through dedicated campaigns. The client's margin rule holds **20 % of the AgLime market open** as operational flexibility: the target curve is calibrated so the annual balance lands at

$$A_{baseline} = 0.8\,A_{cap} = 108\,000\ \mathrm{t/y}, \qquad L_{0/20} = 0 \tag{8.1}$$

The curve is a shape-preserving size rescale of the measured 2026-08-08 belt-cut, $P_{tgt}(x) = P_{meas}(x/k)$, with k found by bisection on the full two-mode annual plan. The result is $k = 1.426$, i.e. a feed whose every characteristic size is 43 % coarser (D_{50}: 32 → 46 mm), delivering

$$\boxed{P_{tgt}(20\ \mathrm{mm}) = 40.1\ \%} \qquad \text{(measured today: 45.5 \%)} \tag{8.2}$$

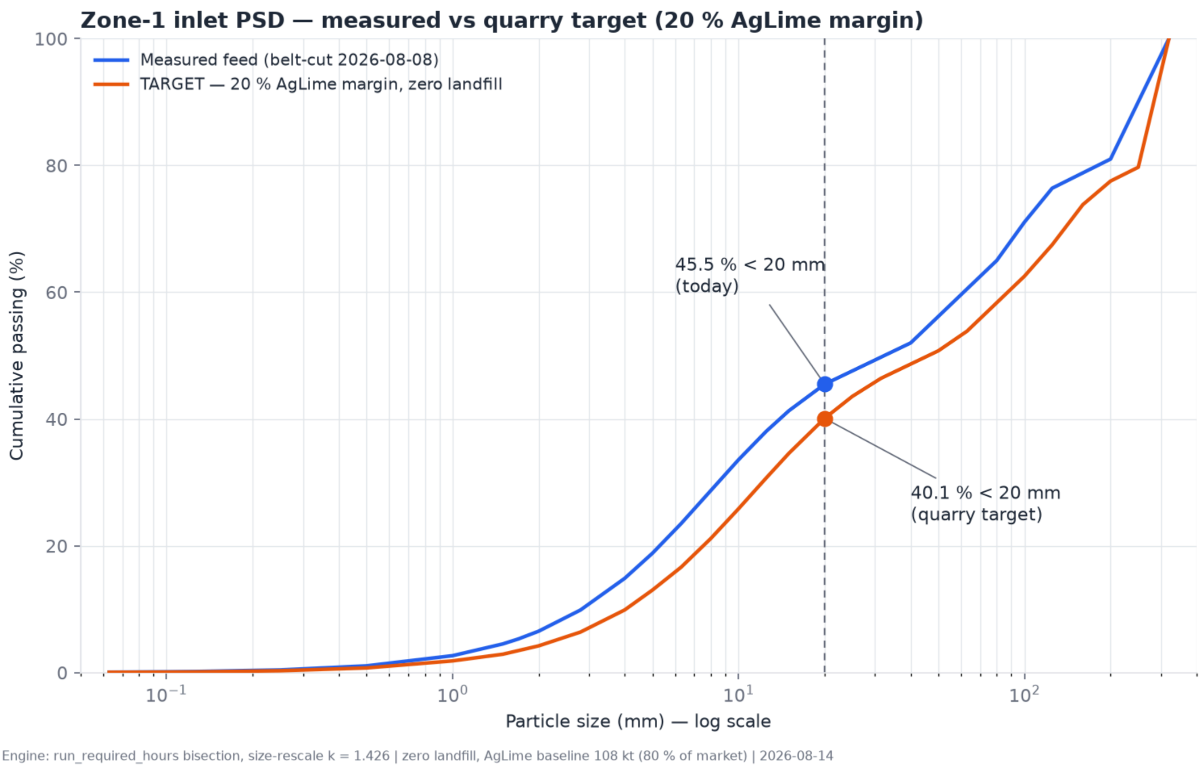

Fig. 8.1 — Zone-1 inlet PSD, semi-log: measured curve vs adopted target. The 20 mm control point carries the whole specification.

8.2 What the target buys

Annual quantity	Today's feed	At target
Quarry extraction for the same firm sales	341 650 t	300 825 t (−40 825)
0/20 to landfill (net loss)	13 816 t	**0**
AgLime baseline / flex reserve	135 000 / 0 t	108 000 / **27 000 t**
KFS Yield realized vs required	24.88 / 25.93 %	28.26 / 28.26 % — self-consistent
Zone utilizations 1.1 / 1.2 / 1.3	71.2 / 40.5 / 60.2 %	62.7 / 36.0 / 60.2 %

The explicit cost of the margin: 27 000 t/y of AgLime sales are not planned in the baseline. They are recoverable at will — whenever the quarry runs finer than target, the excess 0/20 is absorbed by pushing the 2C campaigns toward the full 135 kt market instead of being landfilled. The margin is bought with optional sales, not with waste.

8.3 Control protocol

Step	Rule
Measurement point	belt-cut PSD at the PIVOT (downstream of the primary crusher), standard project protocol
Control point	cumulative passing at **20 mm**
≤ 40.1 %	on target or coarser — full flexibility available
40.1 – 45.5 %	workable — AgLime campaigns absorb the excess; the flex reserve shrinks accordingly
> 45.5 %	worse than today — landfill exposure returns; escalate to the quarry
Live monitor	the engine's KFS Yield alert: each point of gap ≈ 12 kt/y of landfill exposure

> **Top-size caution.** Coarser than target is safe for the balance but not for the machines: the measured feed F_{80} (181 mm) already exceeds the CR.5009 maximum nip size (150 mm — standing engine alert). Top-size control at the primary crusher matters as much as the fines fraction; both belong to the same quarry conversation.

Provenance — every figure in this document is an engine execution: run_scenario / run_required_hours (two-mode plan, 2C campaigns, converged ×2 grid), repository branch wankoe-python-model, 2026-08-14; quarry curve by bisection (scripts archived, replayable without the assistant). Verification dossier: docs/design/zone11-verification (actions 1–6). Decision register: docs/spec-conformity-matrix.md. — **by NOEZYS**.